Chapter 7: Magnetostatics Fields

The following subjects will be discussed

- The Biot-Savart Law.
- Magnetic Field Calculations using Biot-Savart Law.
- Ampere's Law.
- Application of Ampere's Law.

Biot-Savart Law

Magnetostatic fields are generated whenever there are charges moving with constant velocity (DC current).

The basic law for calculating the magnetic field round a current-carrying structure is

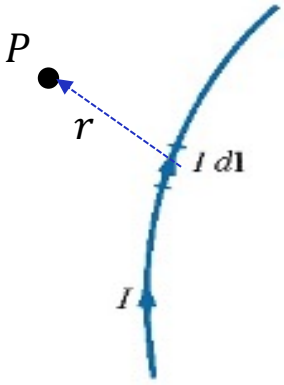
Biot-Savart Law

$$d\vec{H} = \frac{1}{4\pi}\frac{(I\,\vec{dl} \times \vec{a}_r)}{r^2} \qquad d\vec{H} = \frac{1}{4\pi}\frac{(\vec{K}\,ds \times \vec{a}_r)}{r^2} \qquad d\vec{H} = \frac{1}{4\pi}\frac{(\vec{J}\,dv \times \vec{a}_r)}{r^2}$$

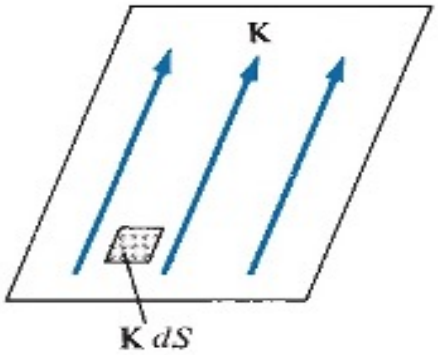
$\vec{H}$: Magnetic field intensity (A/m).

r: Distance to point *P*.

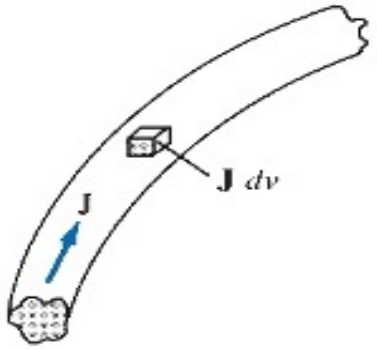
$\vec{a}_r$: Unit length along distance to point *P*. (from wire to point).



Current-carrying Wire



Surface current on a sheet



Volume current in 3D material

Application of Biot-Savart Law

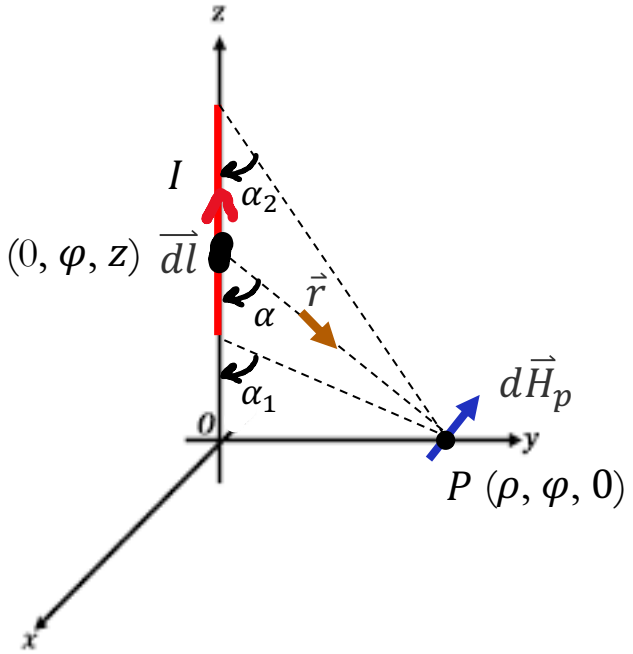
Case (1): Finite constant current-carrying Wire.

Biot-Savart Law $$d\vec{H}_p = \frac{1}{4\pi}\frac{(I\,\overrightarrow{dl}\times\vec{a}_r)}{r^2}$$

$\overrightarrow{dl} = dz\,\vec{a}_z$ $\quad \vec{r} = \rho\vec{a}_\rho - z\,\vec{a}_z$ $\quad r^2 = \rho^2 + z^2$

$$\overrightarrow{dl}\times\vec{a}_r = (dz\,\vec{a}_z)\times\left(\frac{\rho\vec{a}_\rho - z\,\vec{a}_z}{\sqrt{\rho^2+z^2}}\right) = \frac{\rho\,dz}{\sqrt{\rho^2+z^2}}\vec{a}_\varphi$$

$$\vec{H}_p = \frac{I}{4\pi}\int\frac{\rho\,dz}{(\rho^2+z^2)^{3/2}}\vec{a}_\varphi$$



Work with angles. $\quad z = \rho\cot\alpha$ $\quad dz = -\rho\,csc^2\,\alpha\;d\alpha$ $\quad r = \rho\csc\alpha$

$$\vec{H}_p = \frac{-I}{4\pi\rho}\int_{\alpha_1}^{\alpha_2}\sin\alpha\,d\alpha\,\vec{a}_\varphi$$

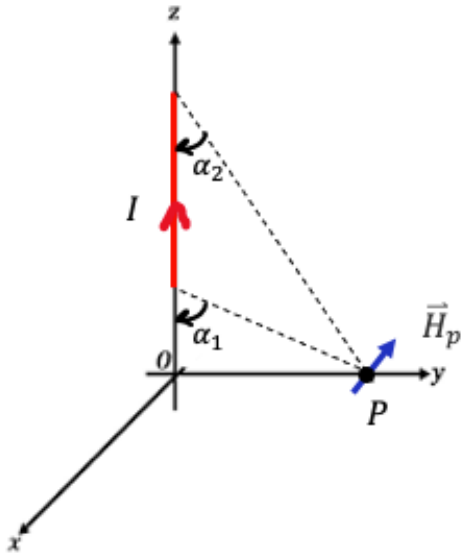
$$\vec{H}_p = \frac{I}{4\pi\,s}(\cos\alpha_2 - \cos\alpha_1)\,\vec{a}_H$$

- s: Normal distance to axis of wire
- α_2: End of wire angle
- α_1: Start of wire angle
- $\vec{a}_H$: By right-hand rule

Application of Biot-Savart Law

Case (1): Finite constant current-carrying Wire.

$$\vec{H}_p = \frac{I}{4\pi\, s}(\cos\alpha_2 - \cos\alpha_1)\,\vec{a}_H$$



A special case: Infinite wire.

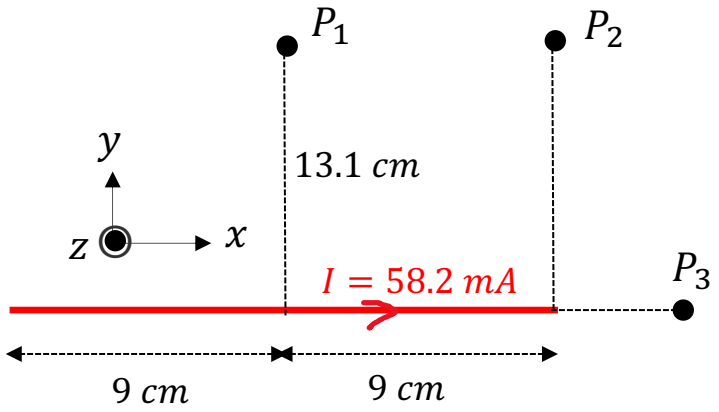
$\alpha_1 = 180°$ $\alpha_2 = 0°$

$$\vec{H} = \frac{I}{2\pi s}\vec{a}_H$$

Example: Find H at P_1.

$\alpha_1 = 124.5°$ $\alpha_2 = 55.5°$ $s = 13.1\ cm$

$\vec{H}_1 = 0.69\ \vec{a}_z$ (A/m)



Find H at P_2.

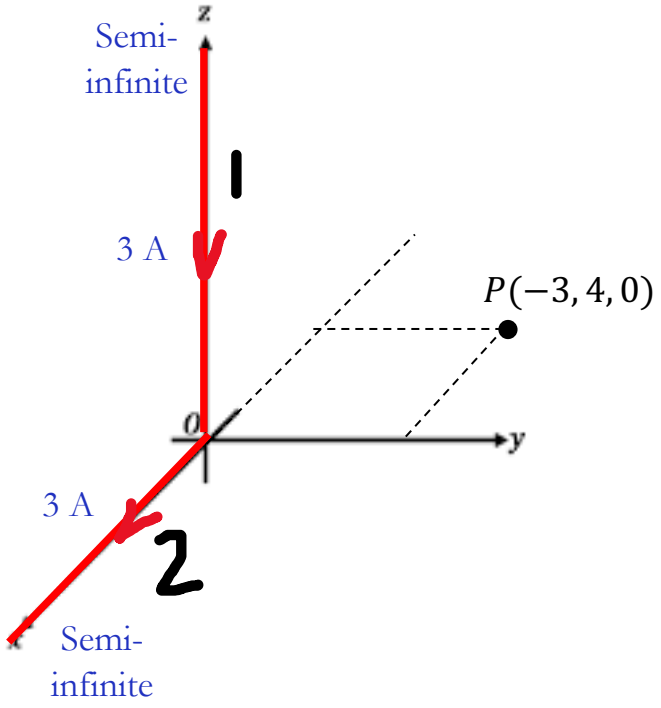
Find H at P_3.

Application of Biot-Savart Law

Case (1): Finite constant current-carrying Wire.

$$\vec{H}_p = \frac{I}{4\pi\, s}(\cos\alpha_2 - \cos\alpha_1)\,\vec{a}_H$$

Example: Find H at P.



For line 1

$\alpha_1 = 180°$ $\alpha_2 = 90°$ $s = 5\,m$

$\vec{a}_{H1} = -\vec{a}_z \times \vec{a}_\rho = -\vec{a}_\varphi$

$$\vec{H}_1 = -\frac{3}{20\pi}\,\vec{a}_\varphi \ \text{(A/m)}$$

For line 2

$\cos\alpha_1 = 3/5$ $\alpha_2 = 0°$ $s = 4\,m$

$\vec{a}_{H2} = \vec{a}_z$

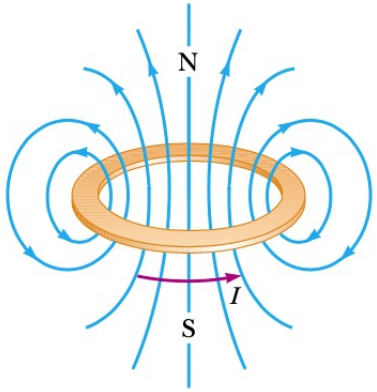
$$\vec{H}_1 = \frac{3}{40\pi}\,\vec{a}_z \ \text{(A/m)}$$

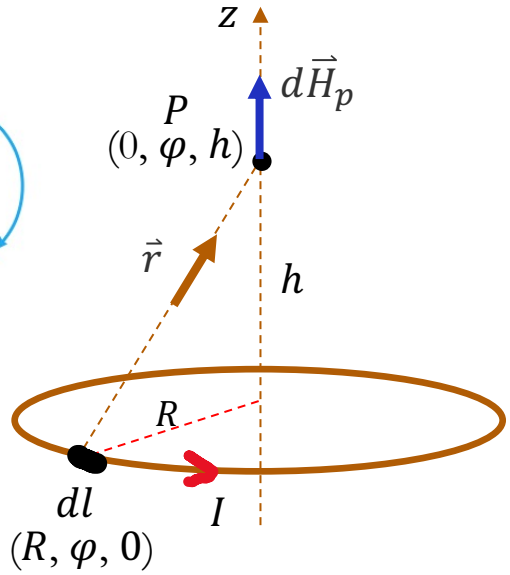
$$\vec{H}_p = \frac{3}{40\pi}(-2\,\vec{a}_\varphi + \vec{a}_z) \quad \text{A/m}$$

Application of Biot-Savart Law

Case (2): Circular Ring (radius R) with constant current. Find $\vec{H}$ along axis

Biot-Savart Law $\quad d\vec{H}_p = \dfrac{1}{4\pi}\dfrac{(I\,\vec{dl}\times\vec{a}_r)}{r^2}$





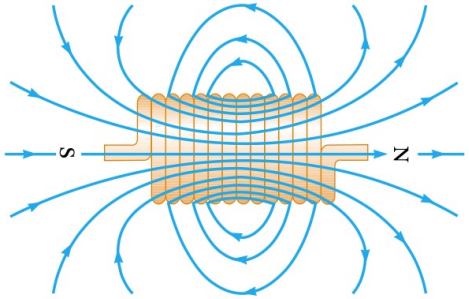
$\vec{dl} = R\,d\varphi\,\vec{a}_\varphi \qquad \vec{r} = -R\vec{a}_\rho + h\,\vec{a}_z$

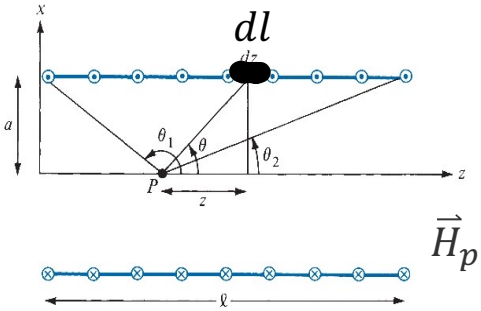
$\vec{dl}\times\vec{r} = Rh\,d\varphi\,\vec{a}_\rho + R^2 d\varphi\,\vec{a}_z$

$$\vec{H}_p = \frac{I}{4\pi}\int_0^{2\pi}\frac{R^2 d\varphi}{(R^2+h^2)^{3/2}}\vec{a}_z \qquad \vec{H}_p = \frac{IR^2}{2(R^2+h^2)^{3/2}}\vec{a}_z$$

Along ring axis

Example: Magnetic field intensity inside a solenoid with turns density $n = N/l$.





$I\,\vec{dl} = n\,I\,dz\,\vec{a}_z \qquad z = a\cot\theta$

$$dz = -a\,csc^2\theta\;d\theta = -\frac{(a^2+z^2)^{3/2}\sin\theta}{a^2}d\theta$$

$$\vec{H}_p = \int_{\theta_1}^{\theta_2}\frac{n\,I\sin\theta\;d\theta}{2}\vec{a}_z = \frac{n\,I}{2}(\cos\theta_2 - \cos\theta_1)\,\vec{a}_z$$

For a long solenoid $(l \gg a)$. $\quad \vec{H}_p = n\,I\,\vec{a}_z$

Ampere's Law

Magnetic fields are generated whenever there is a moving charge (current).

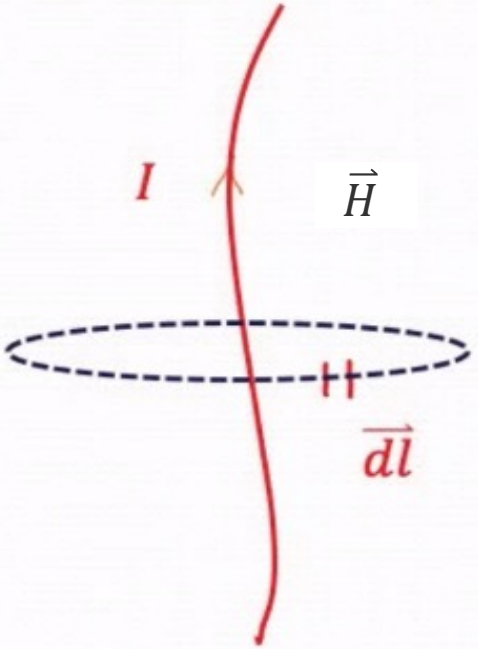
Ampere's Law

$$\oint \vec{H} \cdot \vec{dl} = I_{enclosed}$$

Integration along a closed path

Using Stokes theorem

$$\vec{\nabla} \times \vec{H} = \vec{J}$$



This law can be used to calculate the magnetic field $\vec{H}$ if the current I is known. Only for uniform structures.

An Amperian contour (path).

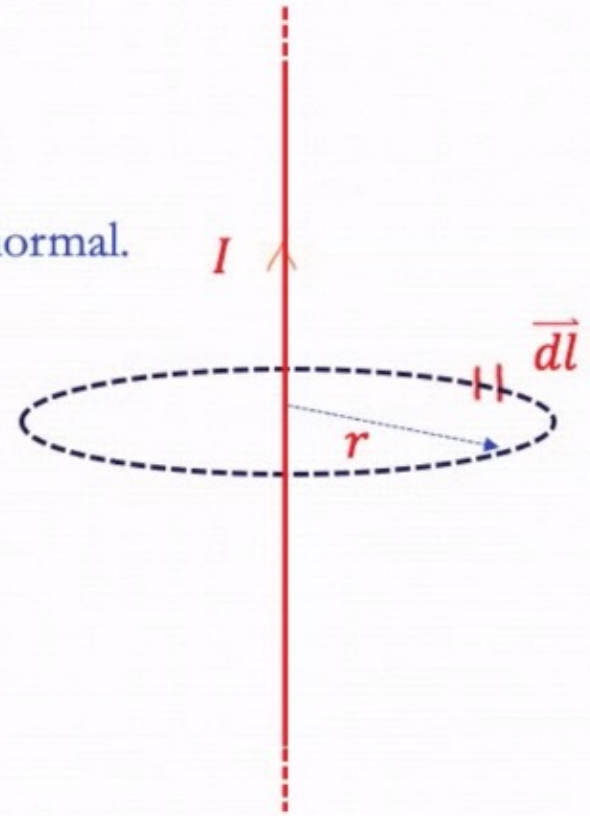
1) Closed
2) Field is constant over path and
3) field is either completely tangential or completely normal.

Application of Ampere's Law

(1) Infinite straight thin wire with constant current I.

A suitable Amperian contour is a circle with radius.

1) Closed
2) Field is constant over path and
3) field is either completely tangential or completely normal.



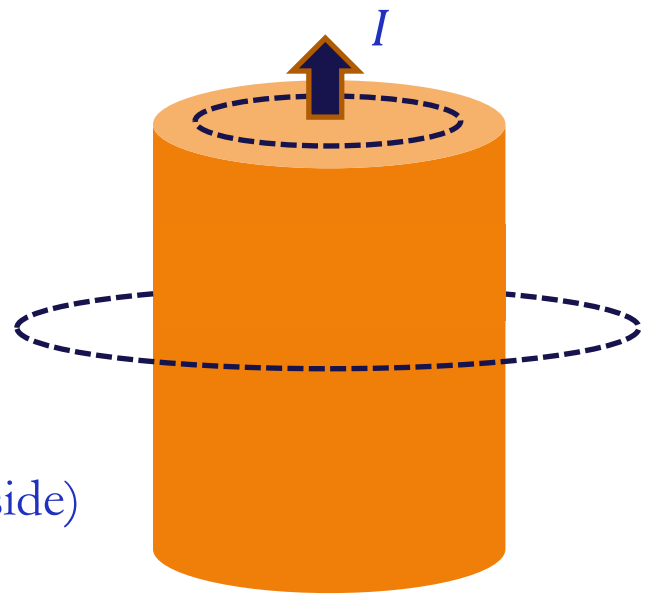
Ampere's Law $\oint \vec{H} \cdot \vec{dl} = I_{enclosed}$

$$\oint \vec{H} \cdot \vec{dl} = H \oint dl = H\,(2\pi r)$$

$$H\,(2\pi r) = I \quad \Rightarrow \quad \vec{H} = \frac{I}{2\pi r}\,\vec{a}_H$$

Application of Ampere's Law

(2): Long straight thick wire with radius R and current I.



A suitable Amperian contour is a circle with radius r.

1) Closed
2) Field is constant over path and
3) field is either completely tangential or completely normal.

Ampere's Law $\oint \vec{H} \cdot \overrightarrow{dl} = I_{enclosed}$

Two regions to consider: (1) $r \leq R$ (inside) (2) $r \geq R$ (outside)

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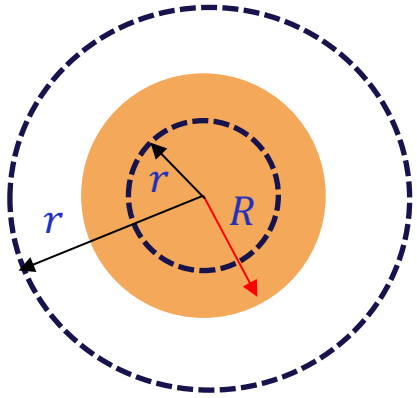
(1) $r \leq R$ (inside) $\oint \vec{H} \cdot \overrightarrow{dl} = 2\pi r H$ $I_{enc} = I \dfrac{\pi r^2}{\pi R^2}$

$2\pi r H = I \dfrac{r^2}{R^2}$ ⇒ $\vec{H} = \dfrac{I}{2\pi R^2} r\, \vec{a}_H$

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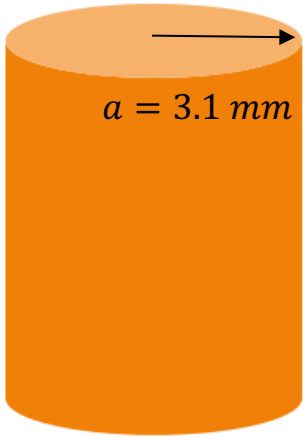
(2) $r \geq R$ (outside) $\oint \vec{H} \cdot \overrightarrow{dl} = 2\pi r H$ $I_{enc} = I$

$2\pi r H = I$ ⇒ $\vec{H} = \dfrac{I}{2\pi r} \vec{a}_H$



Examples

(1) The magnitude of the current density through the wire is given by $J = j_o\, r/a$, where $j_o = 310$ A/m^2. Find the magnitude of the generated magnetic field at $r = 0, 2, a, 2a$. Use Ampere's law.



$r \le R$ (inside) $\quad \oint \vec{H} \cdot \overrightarrow{dl} = 2\pi r H$

$$I_{enc} = \int \vec{J} \cdot \overrightarrow{dA} = \int_0^{2\pi} \int_0^{r} j_o \frac{r}{a}\, r\, dr\, d\varphi = j_o \frac{2\pi}{a} \frac{r^3}{3}$$

$$H = \frac{I_{enc}}{2\pi r} = \frac{j_o}{3a} r^2$$

$r \ge R$ (outside) $\quad \oint \vec{H} \cdot \overrightarrow{dl} = 2\pi r H$

$$I_{enc} = \int_0^{2\pi} \int_0^{a} j_o \frac{r}{a}\, r\, dr\, d\varphi = j_o \frac{2\pi a^2}{3}$$

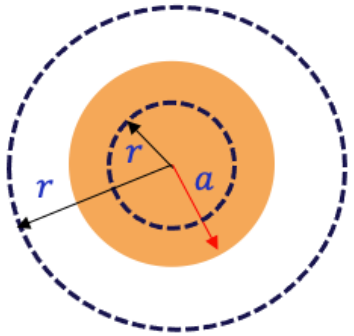
$$H = \frac{I_{enc}}{2\pi r} = \frac{j_o\, a^2}{3} \frac{1}{r}$$

$H(0) = 0$

$H(a/2) = 0.08$ (A/m)

$H(a) = 0.32$ (A/m)

$H(2a) = 0.16$ (A/m)

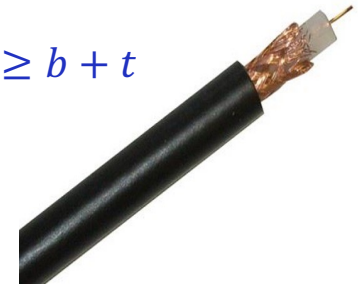


Application of Ampere's Law

(2): Long straight coaxial cable of inner and outer radii of a and b and thickness t. carrying constant current I.

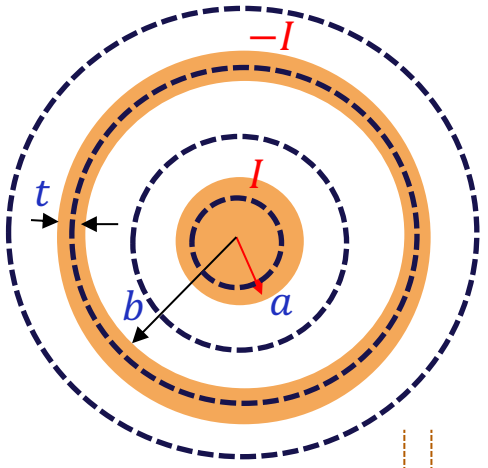
Four regions to consider: (1) $\rho \le a$ (2) $a \le \rho \le b$ (3) $b \le \rho \le b+t$ (4) $\rho \ge b+t$

A suitable Amperian contour is a circle with radius r.



(1) $\rho \le a$ $\quad \oint \vec{H} \cdot \vec{dl} = 2\pi\rho H \qquad I_{enc} = I\dfrac{\pi\rho^2}{\pi a^2} \Rightarrow \vec{H} = \dfrac{I}{2\pi a^2}\rho\,\vec{a}_H$

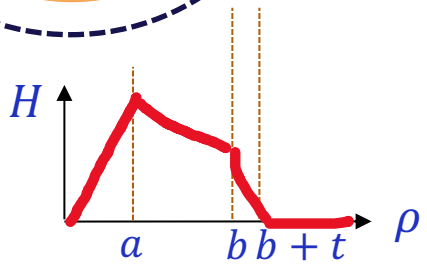
(2) $a \le \rho \le b$ $\quad I_{enc} = I \Rightarrow \vec{H} = \dfrac{I}{2\pi\rho}\vec{a}_H$



(3) $b \le \rho \le b+t$

$$I_{enc} = I\left(1 - \frac{\rho^2 - b^2}{t^2 + 2bt}\right)$$

$$\vec{H} = \frac{I}{2\pi\rho}\left(1 - \frac{\rho^2 - b^2}{t^2 + 2bt}\right)\vec{a}_H$$

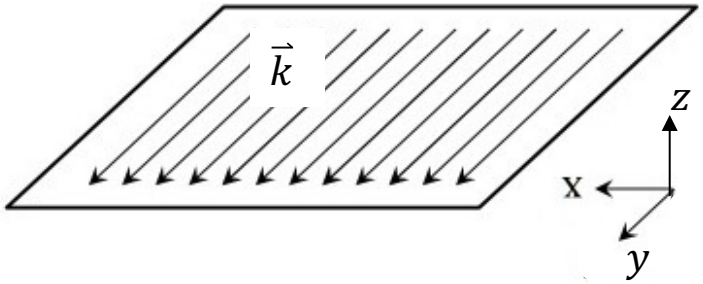


(4) $\rho \ge b+t$ $\quad I_{enc} = I - I = 0 \qquad \vec{H} = 0$

Examples

(2) Using Ampere's law, find the magnetic field generated by an infinite sheet occupying the xy plane and carrying a current density of $\vec{k} = k_o\,\vec{a}_y$ (A/m).

Ampere's law: $\oint \vec{H} \cdot \overrightarrow{dl} = I_{enc}$

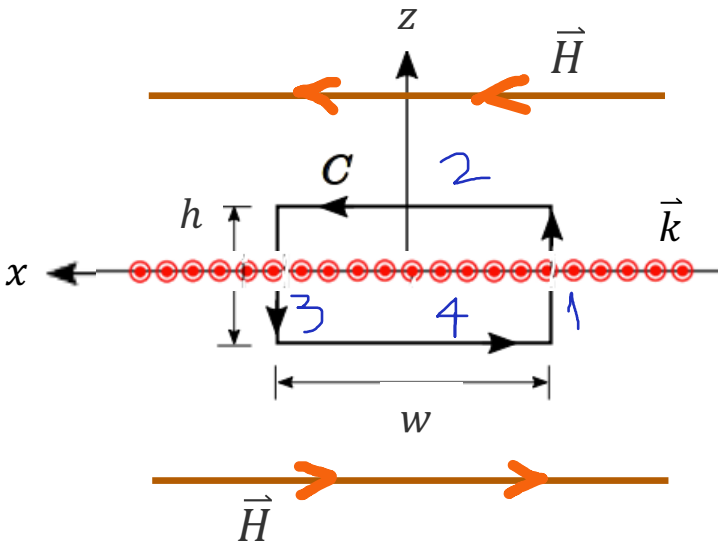


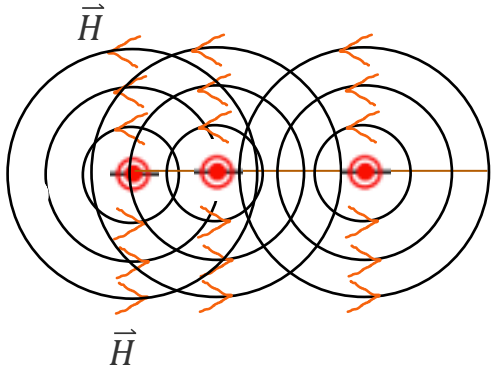
A suitable Amperian contour is a rectangle.

$$\oint \vec{H} \cdot \overrightarrow{dl} = 0 + Hw + 0 + Hw = 2Hw$$

$$I_{enc} = k_o w$$

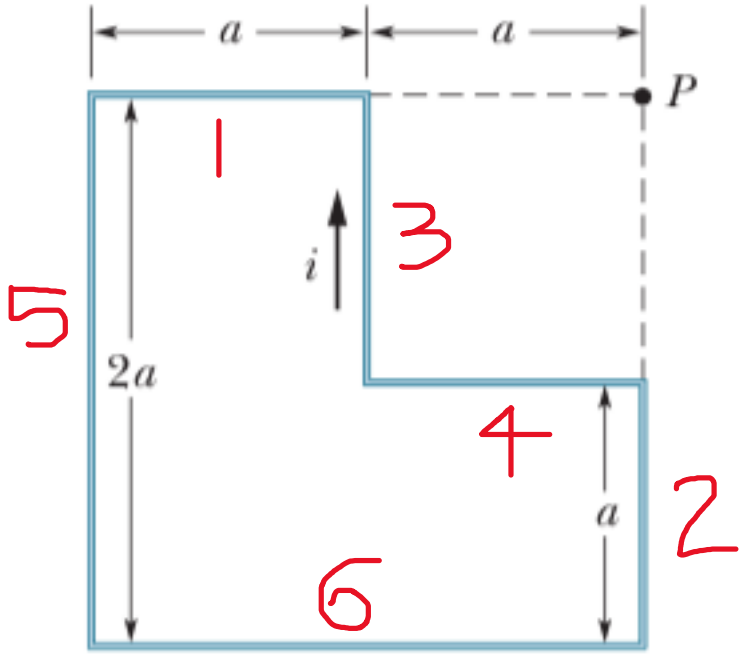
$$\vec{H} = \frac{\vec{k} \times \vec{a}_n}{2}$$





Examples

(1). Find the total magnetic field at point P due to the given structure. The current is constant.



For segments 1 and 2: $H = 0$

For segments 3: $\alpha_1 = 135°, \alpha_2 = 90°$

$$H_3 = \frac{I}{4\pi a}\ \frac{1}{\sqrt{2}}$$ (into the page, +)

For segments 4: $\alpha_1 = 90°, \alpha_2 = 45°$

$$H_4 = \frac{I}{4\pi a}\ \frac{1}{\sqrt{2}}$$ (into the page, +)

For segments 5: $\alpha_1 = 90°, \alpha_2 = 45°$

$$H_5 = \frac{I}{4\pi(2a)}\ \frac{1}{\sqrt{2}}$$ (out of the page, -)

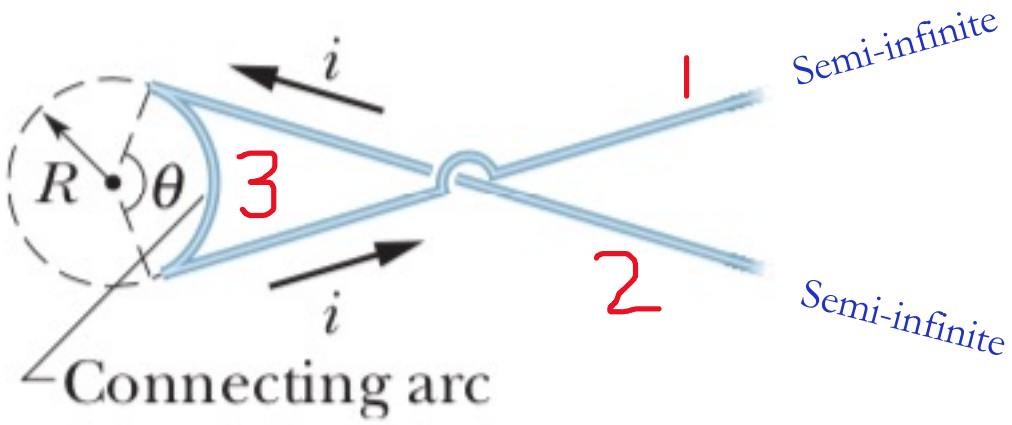
For segments 6: $\alpha_1 = 135°, \alpha_2 = 90°$

$$H_6 = \frac{I}{4\pi(2a)}\ \frac{1}{\sqrt{2}}$$ (out of the page, -)

$$H_P = \frac{I}{4\sqrt{2}\pi a}$$ Into the page

Examples

(2). Find the angle θ to make H equal to zero at the center.



$$H_1 = H_2 = \frac{I}{4\pi R} \text{ (out of the page, +)}$$

$$H_3 = \frac{I}{4\pi R}\theta \text{ (into the page, -)}$$

$$H_{tot} = \frac{2I}{4\pi R} - \frac{I}{4\pi R}\theta = 0$$

$$\theta = 115°$$

Q & A

Send me an email if you have any questions.